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Installation Problems of eSim 2.5 on Ubuntu 25.04

Analysis, Diagnosis and Fixes

eSim Semester Long Internship – Autumn 2026
Screening Task 4: eSim Upgradation

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1 Introduction

eSim is an open-source EDA tool developed by FOSSEE, IIT Bombay. It combines KiCad (schematic capture), Ngspice (analog and mixed-signal simulation), GHDL and Verilator (through NGHDL and NgVeri) and a Python/PyQt5 front end.

The task was to install eSim 2.5 on Ubuntu 25.04, to find the problems that appear during installation, and to fix them by changing the installer scripts. This report lists the problems I found, how I diagnosed each one, the change I made, and how I checked that the change worked.

2 Test Environment

Item	Value
Host virtualisation	Oracle VirtualBox
Guest OS	Ubuntu 25.04 (64-bit)
Python (virtual environment)	3.13
GCC	14.2.0 (Ubuntu 14.2.0-19ubuntu2)
LLVM available by default	20.1.2
KiCad (from Ubuntu repositories)	8.0.x
eSim version tested	2.5 (official zip from esim.fossee.in)
Installer scripts	`installers` branch of `FOSSEE/eSim`

3 Methodology

1. Installed Ubuntu 25.04 in VirtualBox and downloaded eSim 2.5 from the official download page.
2. Ran the installer with all output saved to a log file:

```
./install-eSim.sh --install 2>&1 | tee install.log
```

3. Read the failing line in the log, found the matching code in the installer script, and checked the cause (package names with `apt-cache policy`, syntax with `bash -n`, and the build tool's own `configure` script).
4. Changed the script, checked the syntax, and reran the installer.
5. Repeated until the installer finished.

Structure of the installer. The top-level `install-eSim.sh` only detects the Ubuntu version and then calls a version specific script from `install-eSim-scripts/`. For Ubuntu 25.04 this is `install-eSim-25.04.sh`. That script unpacks `nghdl.zip`, which has its own dispatcher `install-nghdl.sh` and its own version specific scripts in `install-nghdl-scripts/`. Problems therefore appear at three levels: the eSim dispatcher, the eSim 25.04 script, and the NGHDL scripts.

4 Summary of Issues

#	Issue	Component	Effect	Status
1	Syntax error in `install-eSim.sh` on the `installers` branch	eSim dispatcher	Installation never starts	Fixed
2	eSim 2.5 zip has no Ubuntu 25.04 case and no `install-eSim-25.04.sh`	eSim dispatcher	"Unsupported Ubuntu version"	Fixed
3	Version detection returns an empty string on 25.04	eSim dispatcher	Empty version in messages	Fixed
4	`exit 0` inside `installKicad` stops the whole install	eSim 25.04 script	Install silently ends after KiCad	Fixed
5	NGHDL dispatcher has no 25.04 case or script	NGHDL dispatcher	"Unsupported Ubuntu version"	Fixed
6	`libcanberra-gtk-module` not available on Ubuntu 25.04	NGHDL 25.04 script	Install aborts	Fixed
7	GHDL 4.1.0 does not accept LLVM 20	NGHDL (GHDL build)	Build aborts	Fixed

5 Detailed Description of the Issues

5.1 Issue 1: Syntax error in `install-eSim.sh` (installers branch)

Symptom.

```
./install-eSim.sh: line 70: syntax error near unexpected token ‘}’
./install-eSim.sh: line 70: ‘}’
```

Diagnosis. `bash -n install-eSim.sh` reproduced the error with a clean working tree (`git status` showed nothing to commit), so the bug is in the repository itself. Printing the lines around the error with `sed -n ’55,75p’ install-eSim.sh | cat -A` showed that the `if [[ -f "$SCRIPT" ]]` block in `run_version_script` was never closed. Lines from an older installer (`echo "[eSim]" » $config_dir/$config_file` and the start of `installNghdl`) were present in the middle of the dispatcher.

Fix. I replaced the file by a clean dispatcher: the stray lines were removed and the `if` block was closed with an `else` branch and `fi`.

```
    # Run the script if found
    if [[ -f "$SCRIPT" ]]; then
        echo "Running script: $SCRIPT $ARGUMENT"
        bash "$SCRIPT" "$ARGUMENT"
    else
        echo "Installation script not found: $SCRIPT"
        exit 1
    fi
}
```

Verification. `bash -n install-eSim.sh && echo "syntax OK"` prints `syntax OK`.

5.2 Issues 2 and 3: eSim 2.5 zip does not support Ubuntu 25.04, and the version is detected as empty

Symptom. Running the installer from the official zip:

```
dispatcher syntax OK
bash: install-eSim-scripts/install-eSim-25.04.sh: No such file or directory
Detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
```

Diagnosis. Two separate problems:

1. **No 25.04 support (Issue 2).** The `case $VERSION_ID` block lists only 22.04, 23.04 and 24.04, and the zip contains no script for 25.04. The 25.04 script exists only on the `installers` branch of the repository.
2. **Empty version (Issue 3).** The dispatcher obtains the version with

   ```
   FULL_VERSION=$(lsb_release -d | grep -oP '\d+\.\d+\.\d+')
   ```

   The pattern needs three numbers (for example 22.04.4). On Ubuntu 25.04, `lsb_release -d` prints only "Ubuntu 25.04", so nothing matches.

Fix. I copied the dispatcher and the `install-eSim-scripts/` directory from the `installers` branch into the extracted release folder. This adds `install-eSim-25.04.sh` and the `"25.04")` case. The version is read without needing a patch number:

```
FULL_VERSION=$(lsb_release -rs)
```

Verification. The installer prints `Detected Ubuntu Version: 25.04` and then runs `install-eSim-25.04.`

5.3 Issue 4: `exit 0` in `installKicad` ends the whole installation

Symptom. When KiCad is already installed, the installer prints the following and then returns to the prompt without installing anything else:

```
Installing KiCad.........................
Ubuntu 25.04 detected.
KiCad 8.0 is already installed.
```

There is no error message and no further output.

Diagnosis. `grep -n "already installed"` gave line 135, and the next line (136) is `exit 0`. Because `installKicad` is a function called from the main flow, `exit` terminates the *entire script*, so `copyKicadLibrary`, `installNghdl`, `installSky130Pdk` and `createDesktopStartScript` (called after it at lines 395–398) are never run.

Fix. `exit 0` was changed to `return 0`:

```
sed -i '/KiCad 8.0 is already installed/{n;s/exit 0/return 0/}' \
    install-eSim-scripts/install-eSim-25.04.sh
```

Verification. After the change the log continues with `kicad config folder already exists`, `symbol table copied in the directory` and `Installing NGHDL...`.

5.4 Issue 5: NGHDL has no Ubuntu 25.04 support

Symptom. After the KiCad step, the NGHDL installer inside `nghdl.zip` fails with the same message as the eSim dispatcher:

```
Detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
```

Diagnosis. `nghdl/install-nghdl-scripts/` contains only `install-nghdl-22.04.sh`, `install-nghdl-23.0` and `install-nghdl-24.04.sh`. The dispatcher `nghdl/install-nghdl.sh` has the same version regex as the eSim dispatcher and no 25.04 case.

Fix.

1. Created `install-nghdl-25.04.sh` in `nghdl/install-nghdl-scripts/`.
2. Added the 25.04 case to the dispatcher and made the third version number optional (this keeps the existing 22.04.4 check working, which `lsb_release -rs` would break):

```
FULL_VERSION=$(lsb_release -d | grep -oP '\d+\.\d+(\.\d+)?')
...
        "25.04")
            SCRIPT="$SCRIPT_DIR/install-nghdl-25.04.sh"
            ;;
```

3. Repacked `nghdl.zip`, because the eSim installer extracts it again (`unzip -o nghdl.zip`) on every run and would overwrite edits made only in the extracted folder.

Verification. The log shows the 25.04 NGHDL script running and installing its dependencies.

5.5 Issue 6: Package `libcanberra-gtk-module` is not available

Symptom.

```
Installing Gtk Canberra modules.........................
Package libcanberra-gtk-module is not available, but is referred to by another package.
Error: Package 'libcanberra-gtk-module' has no installation candidate
```

Diagnosis. The installer tries to install `libcanberra-gtk-module`, which has no installation candidate on Ubuntu 25.04.

Fix. In `install-nghdl-25.04.sh` the package name was replaced by `libcanberra-gtk3-module`. The older scripts were not changed.

Verification. The step completes and the installer continues to the GHDL build.

5.6 Issue 7: GHDL 4.1.0 does not accept LLVM 20

Symptom.

```
Configuring ghdl-4.1.0 build as per requirements
...
Unhandled version llvm 20.1.2

Error! Kindly resolve above error(s) and try again.
```

Diagnosis. The NGHDL script installs the unversioned LLVM packages and passes the default `llvm-config` to GHDL's `configure` (lines 55, 58 and 105 of `install-nghdl-25.04.sh`):

```
sudo apt remove -y llvm llvm-dev
sudo apt install -y llvm llvm-dev
...
./configure --with-llvm-config=/usr/bin/llvm-config
```

On Ubuntu 25.04 these packages are LLVM 20.1.2, and the `configure` script of GHDL 4.1.0 stops with "Unhandled version".

Fix. Install LLVM 18 explicitly and point `configure` to it:

```
sudo apt install -y llvm-18 llvm-18-dev clang-18
...
./configure --with-llvm-config=/usr/bin/llvm-config-18
```

Verification. GHDL, Verilator and Ngspice were built, and the log ends with `NGHDL installed sucessfully` and `eSim Installed Successfully`.

6 Result

After the fixes, the installer finishes on Ubuntu 25.04 and prints:

```
NGHDL installed sucessfully
Installing SKY130 PDK.....................
-----------------eSim Installed Successfully-----------------
Type "esim" in Terminal to launch it
```

6.1 Installation Screenshots

The following screenshots show the successful installation and the resulting eSim environment.



Figure 1: Installation output showing successful completion.

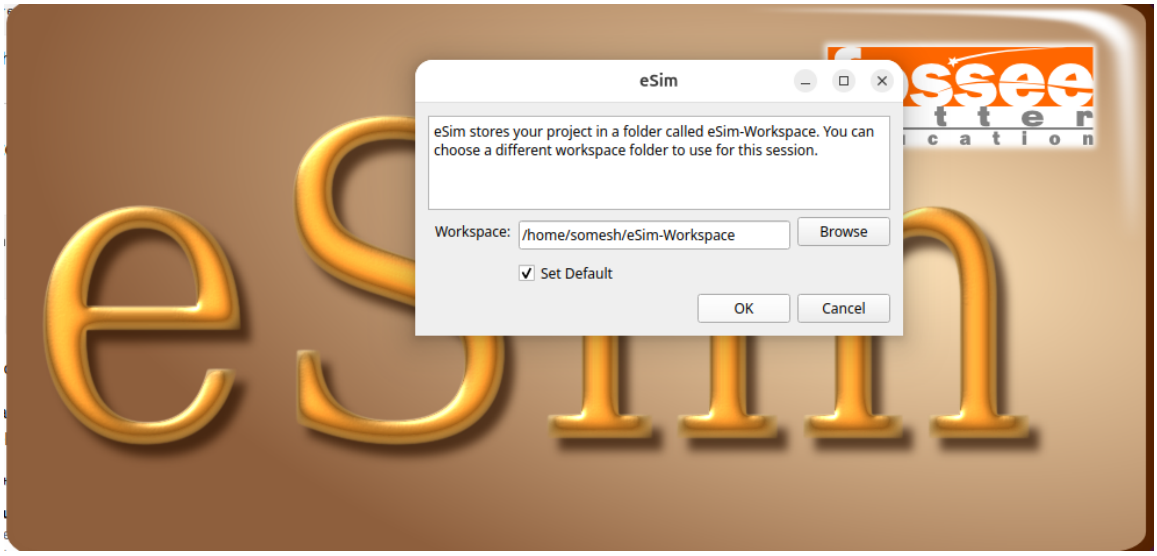


Figure 2: eSim application launch screenshot.

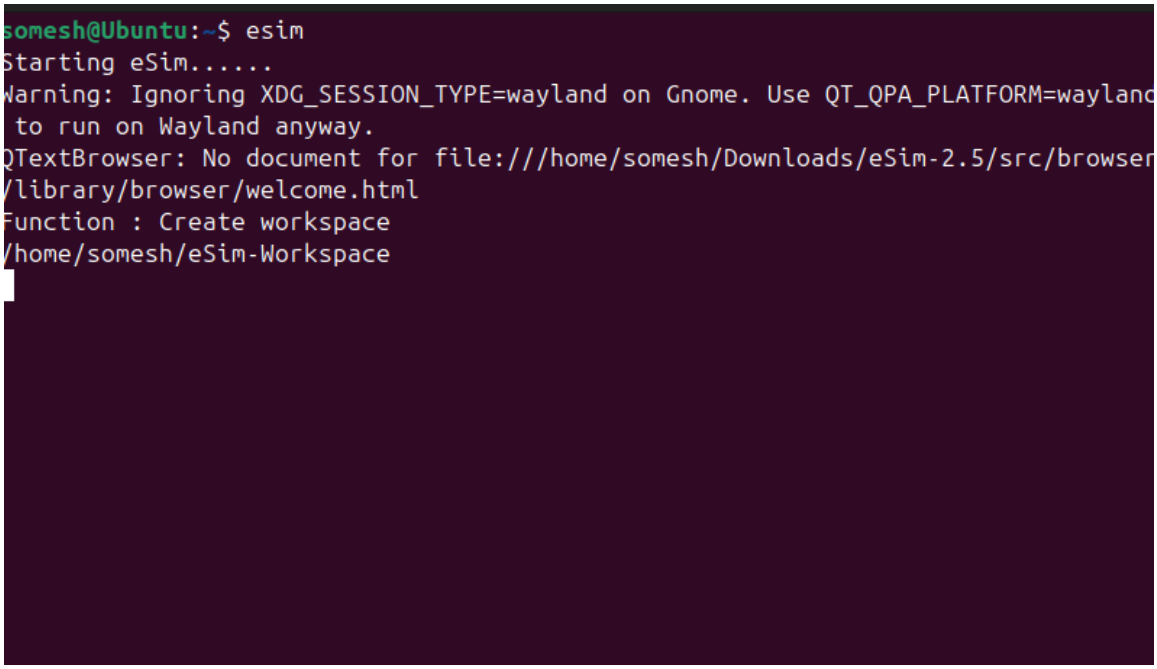


Figure 3: eSim environment screenshot.

7 Files Changed in the Repository

File	Change
`Ubuntu/install-eSim.sh`	Syntax error removed and version detection corrected (Issues 1–3).
`Ubuntu/install-eSim-scripts/install-eSim-25.04.sh`	Changed `exit 0` to `return 0` in `installKicad` (Issue 4).
`Ubuntu/nghdl-fixes/install-nghdl.sh`	Added Ubuntu 25.04 case and corrected version detection (Issue 5).
`Ubuntu/nghdl-fixes/install-nghdl-scripts/install-nghdl-25.04.sh`	Added the Ubuntu 25.04 NGHDL installer with the GTK 3 Canberra module and LLVM 18 fixes (Issues 5–7).

8 Conclusion

Ubuntu 25.04 is not supported by the eSim 2.5 release installer. Seven problems prevented the installation, in the eSim dispatcher, the eSim 25.04 script and the NGHDL installer. All seven were fixed, and the fixes are in the `installers` branch of the repository linked on the first page. After the fixes, the installer completes and reports a successful installation of eSim on Ubuntu 25.04.